

WASP-11b

R-2074

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-10_111023 · created 2026-08-09T23:32:56+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2455857.8191 +/- 0.00025 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.096 +/- 0.021
Transit depth (Rp/Rs)^2	0.92 +/- 0.41 [%]
Orbital Inclination (inc)	87.91 +/- 1.03
Airmass coefficient 1 (a1)	0.964 +/- 0.01
Airmass coefficient 2 (a2)	0.033 +/- 0.023
Scatter in the residuals of the lightcurve fit is	1.05 %
Best Comparison Star	#1 - [67, 58]
Optimal Aperture	3.58
Optimal Annulus	4.46
Transit Duration (day)	0.092 +/- 0.012

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.096 ± 0.021 vs 0.131 (deviation 1.38σ)
Duration fit vs published	0.092 d vs 0.1075 d (deviation 1.07σ)
Mid-time O–C	0.5 min
Depth SNR	3.8
Residual scatter	1.05 %

Light curve

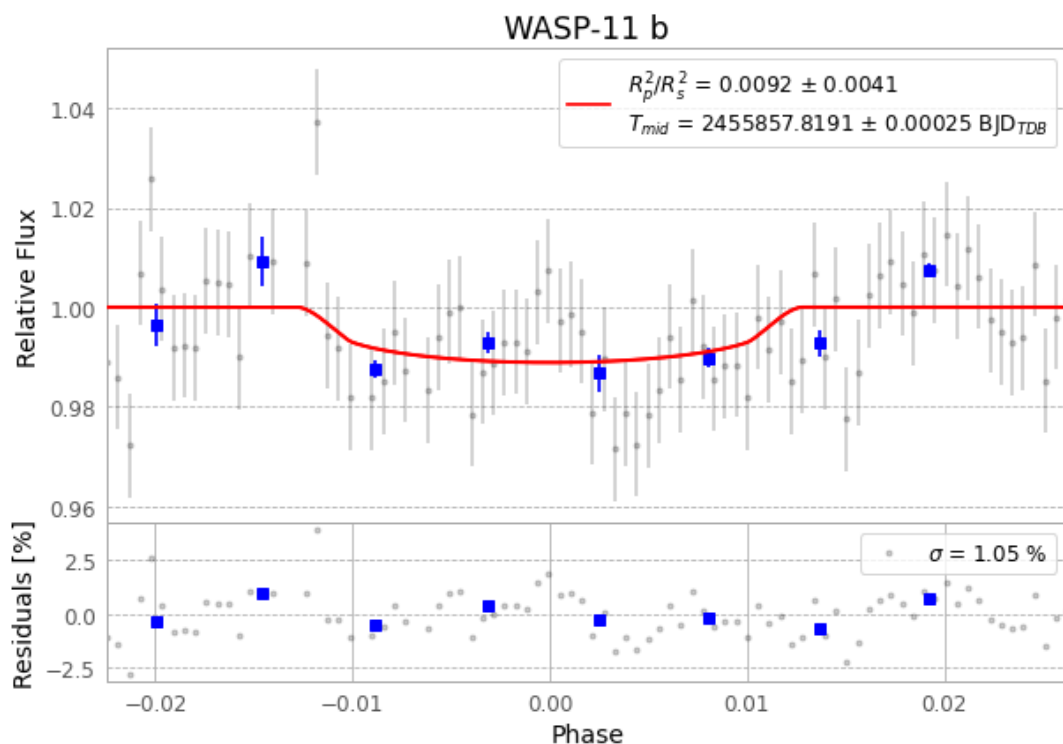


Figure 1 — FinalLightCurve_WASP-11 b_2011-10-22.png (SHA-256 c8703e49ae259933...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [270, 226], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 a85fc705a562e095...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

88 FITS frames (58 MB), HATP-10111023053012.FITS → HATP-10111023095112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-111023.log (f9a6f0ff41c1...), FinalLightCurve_WASP-11 b_2011-10-22.png (c8703e49ae25...), FinalLightCurve_WASP-11 b_2011-10-22.csv (7abf8d0a1a60...)

Compute resources

Wall time 192.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-09 23:32Z.